

Empathy building 'in the wild' - a reflection on an avoidance of the emotional engagement

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ABSTRACT

In 2019 we were asked to prepare a VR stand providing an opportunity to experience applications with social impact during an open event. Every visitor of our stand could choose among several applications (therapeutic, relaxing vs educational vs perspective-taking & empathy-building) and after testing was asked (optionally) to fill in the shortened immersion questionnaire (N=44, age 8 to 62, mean: 31). We have noticed that although people declared they appreciate this kind of experiences, they avoided them (probably because of the potential emotional labor). In this article we share insights and conclusions on differences in conducting research on VR social applications in the controlled laboratory conditions, and in the field testing. Taking into account advantages and disadvantages of this approach as well as methodological challenges, we also advocate for taking this kind of research as a supplementary for typical laboratory research.

Index Terms: Virtual reality—Empathy—Immersion—VRPT; Human-centered design—Field research—On site research

1 INTRODUCTION

Among the mainstream entertainment applications of Virtual Reality, environments designed for educational, training and coworking purposes have recently gained an increasing attention. Among the serious games/apps trend, there are attempts to use the potential of perspective taking, embodiment and immersion offered by Head Mounted Displays (HMD) to present problems of other people or environmental problems.

Virtual reality was recently described as a medium for attitude change - to affect peoples' attitudes, including emotions, behavior, knowledge, and beliefs over an entity [13], especially toward stigmatized groups (like drug use). In the same work, a consciousness of cultural heritage was also investigated.

VR is often referred to as "Empathy machine". There are attempts to use it as a medium to affect human rights attitudes [12] or using immersive storytelling to show problems faced by children in the refugees camps [19]. Facilitating empathy through virtual reality has been a subject of several studies in recent years [22], [23], [25], [26].

Another experience is dedicated to ethnocultural differences and it induces empathy toward out-groups using critical incidents and cultural explanations in the 360 movie [17]. Also racial bias was investigated in [21]. The empathy can be deepened not only to the unfamiliar people, but also to closest ones, e.g. in the mother-child relation [16].

VR seems to be a perfect tool for increasing community consciousness of the environmental crisis and climate change [14], [20].

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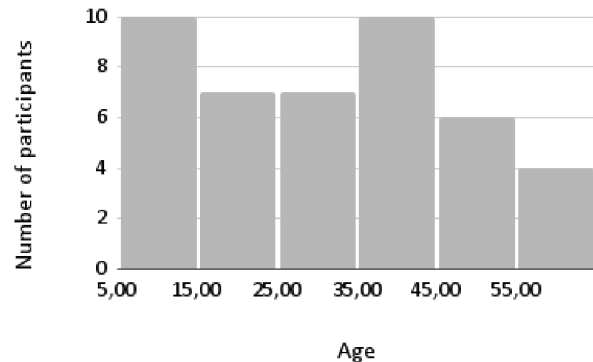


Figure 1: Distribution of participants age.

In [11], Bailenson describes applying the procedure of testing the *Becoming homeless: a human experience* [10] created in his lab to the wider population in the *Empathy at scale* project. The detailed results are reported in [18]. This idea of validation the VRPT experience in the field testing inspired us to design the study reported in this paper.

In June 2019, Poland celebrated the 30th anniversary of events that led to the free elections and on this occasion, Gdańsk city held an event - Celebration of Freedom and Solidarity [4] for the historical memory of those political changes that took place after 1989. As a part of the celebrations, a social zone was prepared for the visitors near the Gdansk shipyard and European Solidarity Centre. The zone consisted of several tents with different non-profit organizations. Our University, as a partner of Open Eyes Economy Summit, was to prepare a VR stand in a Sustainability Tent, which offered trying out VR applications for social impact. Our participation in the event lasted 4 days (1-4 June 2019).

2 MATERIAL AND METHODS

2.1 Participants

Among participants who visited our stand, 44 (21 female and 23 male) volunteered to fill in the immersion form after testing VR experiences. Due to specificity of the event, our stand was visited by families or groups of friends. The average age was 31, but it ranged from 8 to 62 (the age distribution is presented in Fig. 1).

29 % of participants declared that they have had an occasion to use VR before, while for 71 % it was their first experience. As for playing video games, 37,8 % play rarely, % 15,6 - often, 6,7 % every day and 40 % do not use to play at all.

Table 1: Immersion questionnaire. * denotes for an inversed scoring

No	Item
1	To what extent did you feel that you were focused on the application?
2	To what extent have you lost track of time?
3	To what extent have you forgotten about your daily worries?
4	Have you ever felt the need to stop playing and see what was happening around you? *
5	To what extent did you feel that you were interacting with the environment in the app?
6	To what extent did you feel emotionally involved in the app experience?
7	To what extent were you interested in how the events in the app would turn out?
8	To what extent did you like the graphics and artistic vision?
9	Would you like to go through this VR experience again?

2.2 Material

2.2.1 Introductory application

The Lab [5] was usually used as an introductory application to get used to VR experience. For participant for whom it was their first VR experience, The Lab allowed learning the main VR affordances, interaction with controllers and teleportation mechanics.

2.2.2 Perspective taking & empathy building applications

Becoming Homeless [2], [10] is an interactive story in which the attendant experiences main character becoming homeless. It lasts 8 min.

360 movie by The Open Cages foundation is showing cage rearing from a perspective of a hen [1] [7]. Its duration is 2 min.

Notes on Blindness is an experience that visualized the process of losing sight using first person perspective and narration [6]. The experience lasts about 8 min.

2.2.3 Relaxing & therapeutic applications

Nature Treks VR Relaxation through exploring natural landscapes from various climatic zones. It was recently used in a study on therapeutic effect of VR applications [15].

theBlu [8] allows for experiencing the beauty and biodiversity of the ocean.

2.2.4 Educational or creative applications

The Body VR: Journey Inside a Cell is an educational virtual reality experience that takes the user inside the human body [3].

TiltBrush is a 3D artistic content creator [9]. It was also used in the same study on therapeutic effect of VR applications [15].

2.3 The shortened version of Immersion Questionnaire

As a research tool for measuring user experience we used a shortened version of the Immersion Questionnaire. We adapted Polish translation by [24] and shortened to 9 questions (using 5-points Likert scale, from 1-I totally disagree to 5- I totally agree), and replaced word *game* with *application* or *VR experience*.

2.4 Procedure

HTC VIVE Pro with controllers was used in a room scale experience. Majority of experiences were presented in standing position, while for those non-interactive and longer ones, some of the participants were sitting on a chair. An example setup is presented in Fig. 2 and Fig. 3.

Session of each participant lasted about 15-20 min. After short introduction concerning the technology and available experiences the safety aspects, contraindications and potential side effects were explained and the participants signed an informed consent (children



Figure 2: A location for the VR stand.



Figure 3: A setup for VR presentation.

under 18 could try the experience after the consent of their parent or guardian).

At the beginning of the experience, the interaction with controllers and rules of teleportation were carefully explained as well as using the guardian zone as a border of the roomscale. To those participants for whom it was their first VR experience, the introductory part was longer and it started from more passive and static experiences and then we passed to more interactive ones. Usually an introductory part was conducted using Steam experience *theLab*.

The second half of the experience was to be chosen by the participant from the selection presented in section Sect. 2.2. After that, they were asked to fill in the immersion form to assess the application that made the greatest impression on them.

Very often the session ended with a deeper discussion on the potential of VR and existing or possible future applications of this medium in different fields.

3 RESULTS AND DISCUSSION

Choice of the experience. While after first introductory application, the participant could choose their second experience from the presented selection, it happened very often that they were interested and even surprised that such experiences as VRPT for homelessness, blindness or hens conditions exist, but they usually decided that they wanted to try something relaxing or therapeutic. Only about 30 % of our participants tried *Becominghomeless : ahumanexperience*, while nobody picked the blindness experience or a 360 movie of a perspective of a hen in cage rearing (even though the foundation Open Cages had its own stand in the same tent). We assumed that although the participants were aware and appreciated the significance of such applications, they did not want to take this emotional

Table 2: Immersion levels for averaged for each experience

	N=14 theBlu	N=13 Becoming home- less	N=12 Nature Treks	N=2 The Lab	N=2 Tilt Brush	N=1 Body VR	Avg
I	4.48	4.31	4.31	3.94	5.00	5.00	4.38
II	4.39	4.63	4.63	-	3.56	-	4.44
Avg	4.45	4.39	4.39	3.94	4.28	5.00	4.40

labor for themselves to empathize with someone or an animal in difficult conditions or suffering. Probably they were afraid it would be emotionally exhausting or they did not feel comfortable with the emotional effort. These aspects need further exploring in a more systematic research. Also a factor of people around could have significant impact on willingness of choosing such potentially emotionally intensive experiences. Therefore they chose the experience which was emotionally easier or relaxing. Proportion of the selections is presented (*N*) in the Table 2 in the columns headers.

Immersion levels. The reported immersion levels (see Fig. 4) were high for all the experiences. The inter-group statistics comparing the immersion of each applications are not statistically significant due to small and unequal numbers of testers, but we present them in the Table 2. There were also no regularities in the relation between the perceived immersion and participants age and previous VR experience (see Fig. 5).

The heterogeneity of the group Open events provide easier access to reach more diverse group of testers (especially compared to the lab research where usually students of particular faculties take part in the experiments). This allows for exit from the 'bubble' of people of the similar profile. This is for both age of participants (see the age histogram in Fig. 1), their technological competences as well as beliefs and lifestyle. In particular, field testing allows reaching and gaining opinions from people for whom VR is not easily accessible in their everyday life. For some types of lab research, depending on the goals, this heterogeneity is not desirable, and purposeful selection is applied with use of pre-screener, which would be more difficult to process during the field testing.

The effect of new technology and onboarding. For those participants for whom it was their first contact with VR technology (in our case - about 70 %), the effect of new experience and emotional attitude to it could have impacted their declarations in the questionnaire. Their attitude could also be influenced by positive or negative opinion shaped by information from media. In the onboarding process, the digital competences (correlated with playing video games) are particularly meaningful. We have observed that children and regular gamers start interacting in VR more easily. The onboarding process is very demanding on the reserchers - they should be empathetic, of good manners and adjust the rate and detail of provided support to each tester individually.

Previous VR experience. If a tester has already tried VR technology before, then the attitude to the presented apps (or their willingness to take part in the testing) could be influenced by the quality of the former experience. We sometimes met people who were discouraged to the VR technology in general because they experienced simulator sickness in the past (usually with the older generations of headsets) or improper application selection (e.g. trying rollercoasters as their first experience).

Impact of observers. While in the lab experiment usually the only observers are the researchers, during the field testing, a participant is aware that their companions (family or friends) or anyone from the event can observe their performance. Sometimes it plays an integrating role (people giving cues to the tester) but it

can also make the tester less natural and afraid if they will be able to cope with the task or - especially in VR case - they were sometimes afraid of looking and behaving weird while wearing HMD.

Dynamics of the experiments The research process in the lab is usually highly orderly and carefully planned, while during open events some spontaneous disruptions are highly probable and researchers must be flexible to various circumstances and ready to adjust the procedure by improvising. This makes the control over the experiment more difficult and the results less regular and less interpretable, especially in the qualitative approach. It happens more frequently that respondents decide not to continue the procedure (sometimes for simple reason that they hurry up).

Challenges for researchers. The work of researchers conducting the field testing is usually more exhausting than during the lab experiments, due to more irregular procedure, the necessity to be more flexible and dynamically adapt to different testers (with various background, competences, attitude). The cognitive load is also related to the dynamics of the experiment and higher emotional arousal of the situation. Both the irregularity of the procedure and a set of additional aspects which should be taken into account by researchers is discussed also by Bailenson [11] while reporting the *Empathyatscale* project.

Technical conditions Also the circumstances of the test are more dynamic than in the lab. Irregular background noise is more probable to induce break-in presence effect. There is higher probability that some unexpected factors interfere with the study.

Attitude of the participants Contrary to the lab conditions, field testing has the advantage of more informal meeting atmosphere. Participant are usually less tense, they let themselves to be more expressive than during following the procedure in the lab. On the other hand, in the lab the participants are usually more focused. That does not mean taking the experiment less seriously if placed during on open event - usually they do their best to apply to the procedure, but treating it more like a part of entertainment than a rigid task. This effect is particularly visible at the end of the experiment, during the contextual interview, when they usually were more relaxed and enthusiastic. At the same time, in the field testing the participants are usually more impatient and they have less time for a session or - particularly for filling in the forms. This was the reason for shortening the immersion questionnaire to 9 items (in the pilot study, first testers complained it was too long). This aspect limits the opportunity to apply larger battery of questionnaires - people could have tendency to avoid them or fill them in carelessly.

COVID-19. The described research took part about a half a year before the beginning of COVID-19 pandemic. Since then, the circumstances of conducting such open events have diametrically changed, starting from the necessity of using disinfectants e.g. antiviral fluids for hands as well as hygienic headset and controllers covers, collecting COVID questionnaire, measuring temperature, and ending with less attendance to the event or even less willingness to take part in the study or higher distrust to the contact caused by fear of COVID infection. Additionally, applying face masks makes it less comfortable to use HMDs.

Final remarks. The insights described above are consistent with our experience from conducting similar presentation and workshops during other festivals 2017-2020 (e.g. Festival of Science and Art, Open Day at University, Scientists Night), where we presented VR applications to over 500 participants. Nevertheless, the numerical results presented in this study are not statistically significant and should be interpreted only qualitatively. Also, the experiences were of different lengths and quality, which also makes it not reliable to compare them directly.

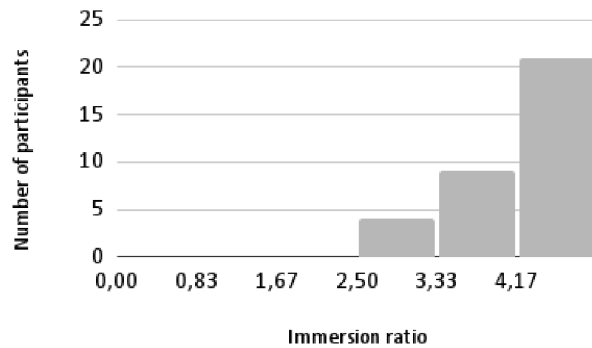


Figure 4: Average immersion levels.

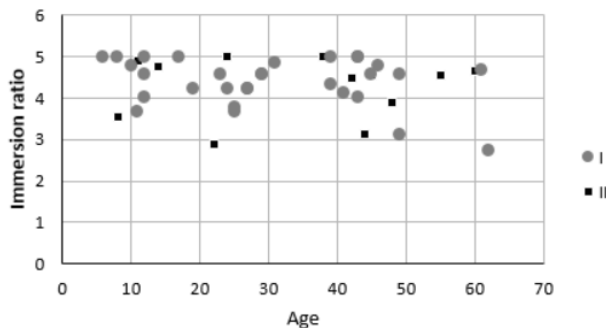


Figure 5: Immersion levels depending on participant age and group (I - first contact with VR, II - tried VR before).

4 CONCLUSION

In this paper we report the observations made during conducting VR research on applications with social impact during an open festival. On the basis of the experience, we pose several open-ended questions on methodological issues and applicability of results of this kind of field research.

Depending on the previous experiences with VR and attitude towards VR they formed in minds before they attend the specific open festival VR exhibition - the whole variety of only partially aforementioned initial factors, but also the absolute necessity of the highest possible attendant support, highly performant hardware for smooth and comfortable VR experiences, interview-based VR sessions experiences cautiously selected for highest possible user experience customized case-by-case, fluent and undisturbed both-sided attendant-supporter communication, reducing noise pollution and potentially stressful distractors to possible minimum - all of these, each more or less, are crucial determinants of the impressions resulting from VR experiences during open festivals.

Taking into account both advantages and shortcomings of field testing, we assume that this form is perfectly supplementary to the lab testing and it is worth this additional effort to reach more diverse group respondents and gain valuable feedback. Especially, applications designed to help others and change their perspective of thinking, in order to validate their impact and make real influence on human attitude, should be made available to the widest group possible.

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